

achieved, the projects disintegrate. In some instances, a new, related project can take over the task of a completed one.

Each of these projects has its own special features but the process of participation is almost the same. A person interested in becoming a user downloads an application from the project website. After installation, the application contacts the respective project's control node, which then transmits a piece of data to the user's computer for analysis. The software analyses the data, driven by unutilised CPU resources.

It is very likely that the PC that you are using is not utilising the total computing power at a given time. Even when you are engaged in a game, creating graphics or surfing the Web, you may only be employing a fraction of your machine's available computing power.

The project's software generally has a very low resource priority; if the user needs to activate a program that needs a large amount of processing power, the project software switches off temporarily. Once CPU usage returns to normal, the software begins analysing data again. Ultimately, the user's computer finishes the data analysis asked for.

At that time, the project software sends data back to the control node, which relays it to the relevant database. Then, the control node sends a new set of data to the user's computer, and the cycle replicates itself. If the project garners sufficient applicants, it can meet ambitious goals in a relatively short span of time.

Privacy and security aspects.

Most grid computing firms provide an assurance that the applicant's

privacy will not be overrun in any way. The software is allowed to only update project-specific data in its own files. Of course, members always have the option of opting out of a project, at any point of time. Applicants can generally decide what level of their system resources

on the company website and on any scoreboards.

Endpoint

For grid to enter the technology mainstream, it will need to move beyond computer-intensive applications that are parallel in nature, to transactional applications like online banking and typical business functions.

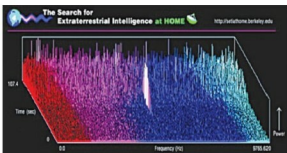
The way applications are developed will also have to evolve for the grid to gain greater approval. Grid computing is being talked of as the next big thing for corporate computing, and it can enable companies to make more optimal use of their IT.

The biggest advantage of grid computing for users is that they can run multiple applications without modification, rather than having to rely on dedicated computer clusters in the datacenter. A user's application will request a Web service, and some computer on the grid will respond.

The grid could also be used in cities to provide applications with higher flexibility, as there would be no single point of failure. Grid computing could become the most preferred method for users to access the data-

base. However, one issue yet to be resolved is licensing, as there is no way the model for licensing today can support on-demand computing.

As grid computing systems' complexity increases, it is likely to result in more organisations and corporations developing multipurpose networks. We can even look forward to a time when corporations inter-network with other companies. In that environment, computational problems that seemed impossible may be condensed to a project that has the duration of a few hours. Let us wait and watch! **EFY**



A SETI packet processing in progress (Image courtesy: www.omniasecuritas.com)



EGEE grid links more than 300 computing centres in more than 50 countries worldwide. The figure shows the European region (Image courtesy: www.kit.edu)

they would like the grid computing company to use. They can also give their preferences in terms of whether the program runs as a screensaver or an application, when computation and communication can be done, whether connections should be made automatically and which proxies and firewall settings to use and so on.

To safeguard member privacy, members are made to create a member name when they install an agent. This name is stored in the member profile directory and used to identify each member